SENTINEL — Privacy-First Local Multimodal AI Companion

Official Submission Brief • Qualcomm Snapdragon AI Lab Build & Present Challenge

One-Line Description

A privacy-first local multimodal AI companion that sees, reads, hears, remembers, understands, and helps users find information across their workspace.

1. Problem

Knowledge workers navigate a fragmented landscape where AI tools lack physical context:

- **Modality Fragmentation**: Vision tools lack memory; chat models cannot see desk hardware; voice transcription lacks spatial awareness; notes lack contextual retrieval.
- **Cloud Privacy Risk**: Sending live desk camera feeds, confidential documents, or proprietary hardware schematics to third-party cloud APIs can create significant data privacy and compliance risks.
- **Physical Hallucination**: Cloud models without grounded visual evidence frequently fabricate object positions and invent document citations.
- **Network Dependency**: Cloud-dependent workflows introduce latency and require connectivity, limiting interactive spatial search and offline use.

2. Solution

SENTINEL is a privacy-first, local-first AI companion that coordinates physical workspace perception and local semantic memory into one unified multimodal loop. Verified core components — vision, OCR, speech, semantic memory — run on-device with no external network calls. A Qualcomm Snapdragon QNN/QAIRT/GenieX runtime adapter is prepared for deployment on Snapdragon X-series hardware.

3. Core Workflow

SEE → **READ** → **HEAR** → **REMEMBER** → **UNDERSTAND** → **FIND**

4. Key Capabilities

- **Object Detection & Tracking (SEE)**: YOLO-X Small ONNX; IoU/centroid tracker with EMA smoothing; 9 normalized spatial zones. Verified on host CPU.
- **Document OCR (READ)**: EasyOCR CRAFT + CRNN with deterministic reading order and spatial text binding. Verified on host CPU.
- **Local Speech-to-Text (HEAR)**: Whisper-Tiny push-to-talk; RTF 0.264 on host CPU (>3.7× real-time). Zero cloud audio.
- **Semantic Memory (REMEMBER)**: all-MiniLM-L6-v2; 384-d hybrid cosine+lexical search; SQLite storage. Verified on host CPU.

- **Multimodal Orchestration (UNDERSTAND)**: SentinelOrchestrator with evidence collection, token budgeting, ResponseValidator. Verified on host CPU.
- **Spatial Find It (FIND)**: Natural-language object localization returning HIGHLIGHT_OBJECT visual grounding actions. Verified on host CPU.

5. Privacy by Design

- No cloud AI API dependencies for any core capability.
- Camera and microphone OFF by default; both require explicit user activation.
- Zero raw video frames or audio recordings saved to disk by default.
- No facial recognition, no biometric profiling, no ambient listening.
- User-controlled memory: knowledge cards saved only on explicit user command.

6. Qualcomm Snapdragon Relevance

SENTINEL is architected for Qualcomm Snapdragon X-series Copilot+ PC deployment:

- **Target Platform**: Snapdragon X Elite / X Plus (Windows 11 ARM64)
- **Target Runtime**: Qualcomm AI Engine Direct (QNN), QAIRT, GenieX
- **Runtime Adapters**: Decoupled CPURuntimeAdapter ↔ QNNRuntimeAdapter — execution provider swappable without application changes.
- **VLM Target**: Qwen3-VL-4B-Instruct via Qualcomm GenieX (adapter staged, awaiting hardware validation)

Hardware Truthfulness Statement

Current development host: Intel Core i5-12450H (x86_64, Windows 11). Physical Snapdragon NPU execution has NOT been verified on hardware. This is disclosed truthfully across all application interfaces, documentation, and pitch materials. No fabricated Snapdragon performance metrics appear anywhere in this submission.

7. Technical Architecture

Browser/Next.js UI → FastAPI → SentinelOrchestrator → [VisionEngine | OCREngine | SpeechEngine | MemoryRetriever] → Runtime Adapter Layer → SQLite + Local Models ↔ Qualcomm QNN Target

8. Verified Benchmarks — Intel Host CPU

All measurements are empirical, taken on the development host (Intel Core i5-12450H, CPUExecutionProvider):

- **YOLO-X inference latency**: 88.34 ms (11.32 FPS theoretical)
- **Whisper-Tiny (3.0s audio)**: 792.9 ms — RTF = 0.264 (>3.7× faster than real-time)
- **MiniLM query embedding**: 7.89 ms (273.1 items/sec batch throughput)
- **Orchestrator overhead**: 26.2 ms (outside VLM inference)
- **Backend test suite**: 145 passed, 0 failed
- **TypeScript errors**: 0 errors

- **Production build**: PASS — 8 static routes

9. Limitations

- Snapdragon NPU execution NOT yet verified — requires physical Snapdragon X-series Windows ARM64 hardware.
- Full VLM inference (Qwen3-VL-4B, 4B params) requires NPU or discrete GPU; runtime adapter staged for target.
- Small-model accuracy limitations in extreme lighting or domain-specific vocabulary.
- EasyOCR and YOLO-X are constrained by host CPU throughput on development machine.

10. Reproducibility

No cloud accounts or API keys required. After initial model artifact download via Hugging Face cache / ONNX model registry, SENTINEL operates fully offline. One-click Windows launcher: run_sentinel.bat. Pre-flight verifier: python scripts/verify_installation.py.